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AI and Transformative Learning in Higher Education: A Systematic Literature Review and Bibliometric Insights

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Abstract

This study comprehensively maps the development and trends in AI and transformative learning research in higher education from 2019 to 2025. Using a systematic literature review and bibliometric analysis, it answers six key questions to explore the evolution of AI integration in transformative learning. Analyzing 181 Scopus-indexed articles, the study utilizes R-studio and VOSviewer software, following the PRISMA method to assess author collaborations, theme evolution, and publication distribution. The results show a significant increase in publications on AI and transformative learning. Initially focused on general AI applications in education, research has shifted toward more specific themes like generative AI, personalized learning, and ChatGPT. Despite technological innovation, pedagogical studies on transformative learning, such as active and personalized learning, remain underexplored in AI contexts. The research also reveals that countries like India and Indonesia dominate the field, indicating regional research concentration. While AI shows potential to improve student motivation, writing skills, and personalized learning, challenges such as ethical concerns, digital literacy, and socio-cultural sensitivity persist, especially regarding academic integrity and AI dependence, which may reduce critical thinking and metacognitive reflection essential for transformative learning. This study affirms that AI must be integrated with a human-centered approach to support both learning effectiveness and critical reflection. Thus, the development of ethical frameworks, educator training, and international collaboration is crucial for the sustainable and inclusive implementation of AI in higher education. In conclusion, while AI offers significant potential for enhancing transformative learning, its successful integration into higher education requires careful consideration of ethical, pedagogical, and socio-cultural dimensions to ensure its responsible and impactful application.

Introduction

The massive development of artificial intelligence (AI) technologies in the past decade has brought about significant transformations in higher education ecosystems around the world (Cantú-Ortiz et al., 2020a; Katsamakas et al., 2024; Okunlaya et al., 2022a). AI is no longer just an experimental project, but has been widely integrated in various aspects of the learning process in higher education, from intelligent tutoring systems, automated assessment, personalized learning, to the utilization of analytics for decision-making and academic support (George & Wooden, 2023a; Lin et al., 2023). Recent studies show that more than 80% of students and faculty in various countries have utilized AI-based tools to support academic writing, research, and collaborative learning (Msambwa et al., 2025; Song & Song, 2023). This massive adoption is driven by AI's ability to facilitate adaptive learning pathways, automate routine tasks, and provide real-time data-driven feedback, which is believed to improve the effectiveness and efficiency of learning in higher education (Simbolon et al., 2025; Song et al., 2024; Strielkowski et al., 2025a).

On the other hand, transformative learning is gaining attention as an important paradigm in higher education (Milazzo & Soulard, 2024; Moyer & Sinclair, 2020). Rooted in Mezirow's theory of transformation of perspective, transformative learning emphasizes the importance of critical reflection that encourages learners to review basic assumptions, reconstruct perspectives, and develop adaptive capacity to deal with the complexity and dynamics of global change (Henukh et al., 2024; Saliba, 2024). In line with the demands of the 21st century, universities are now required not only to produce graduates who are academically competent, but also able to think critically, learn throughout life, and have social responsibility (Pouru-Mikkola & Wilenius, 2021).

The contextualization of transformative learning in today's digital age points to the convergence between AI and transformative learning as a key driver of educational innovation (Cox, 2021; Iyer-Raniga & Andamon, 2016). Recent research has shown that the use of AI can strengthen the transformative learning process by supporting metacognitive reflection, facilitating collaboration in learning, and providing personalized feedback that challenges learners' conventional thinking (Gunnlaugson et al., 2023) . The integration of AI in transformative learning environments is also believed to promote inclusivity, bridge access gaps, and accommodate the diverse needs and backgrounds of students (Gunnlaugson et al., 2023; Morris, 2020). However, these developments also pose new challenges, particularly in relation to the ethical use of AI, digital literacy, and the changing role of educators as facilitators of change that is both technological and transformative (Pedro et al., 2019a; Zhang & Zhang, 2024) .

The urgency to critically examine the relationship between AI and transformative learning is becoming increasingly apparent, especially in the midst of accelerating digital transformation and post-pandemic educational paradigm shifts (Leal Filho et al., 2024) . Therefore, a systematic and bibliometric review is needed to map research trends, identify key themes, gaps, and future development directions in this field (Balaid et al., 2016; Mishra et al., 2024). Through systematic literature review and bibliometric analysis, this article aims to provide a comprehensive overview of how AI shapes transformative learning in higher education, as well as offer theoretical foundations and practical implications for researchers, practitioners, and policy makers in the field of education (Chasokela, 2025; Lytras et al., 2024; Okunlaya et al., 2022b).

In line with the widespread integration of artificial intelligence (AI) technology in various aspects of higher education and the increasing attention to transformative learning as an important paradigm in developing students' critical thinking and adaptive capacity, this study has the main objective to comprehensively explore how AI plays a role in supporting the transformative learning process in higher education (Ghnemat et al., 2022; Imran et al., 2024; Lytras et al., 2024). Through a systematic literature review approach and bibliometric analysis, this study seeks to identify key roles of AI in facilitating transformative learning, map research trends and developments related to the integration of AI and transformative learning, and explore key themes, researcher collaborations, and publication patterns in this area.

Based on this background, this research aims to analyzes relevant data from previous research to answer the following research questions (RQ):

RQ1. What are the descriptive features of the retrieved empirical studies and thematic evolution on AI and transformative learning?

RQ2. What are the summary of related systematic literature reviews in AI and transformative learning?

RQ3. What are the most relevant high-impact sources through Bradford’s Law of Scattering in the field of AI and transformative learning?

RQ4. What are the most cited articles on AI and transformative learning?

RQ5. What are the most relevant countries for AI and transformative learning?

RQ6. What are the trending research themes for future investigations in the field of AI and transformative learning?

Literature Review

Artificial intelligence in higher education

Artificial Intelligence (AI) has become one of the central topics in the transformation of higher education in the digital era (Cantú-Ortiz et al., 2020b; Mohamed Hashim et al., 2022). Simply put, AI is defined as the ability of computer systems to perform tasks that typically require human intelligence, such as understanding language, solving problems, making decisions, and learning from experience (Alkatheiri, 2022) . In education, the presence of AI does not just bring new technology, but also creates a new paradigm in how to manage learning, assess competencies, and facilitate the needs of learners in a more personalized and adaptive manner (Sajja & Sajja, 2021).

The theoretical foundation for the use of AI in higher education can be traced to the concept of constructivism, where learning is seen as an active process involving direct experience, reflection, and adaptation to individual needs (Almulla, 2023; Dahl & Mørch, 2025). AI has the ability to personalize learning experiences according to students' characteristics, interests, and ability levels, making the learning process more relevant and meaningful (Chen et al., 2020). By utilizing learning behaviour data and analytics, AI-based systems are able to recommend learning resources, adjust the level of difficulty of the material, and provide real-time feedback

(Almuhanna, 2025; Murtaza et al., 2022). A number of studies have confirmed the positive contribution of AI in the higher education environment. AI-based adaptive learning systems can analyze learning styles, achievement levels, and obstacles faced by individual students (Strielkowski et al., 2025). This enables the development of a more personalized learning path, where students are given materials, exercises, and evaluations according to their individual needs (Ezzaim et al., 2024)

The previous research shows that students who learn through AI-based adaptive learning platforms show improved concept understanding, learning motivation, and better retention of material, especially in science and technology courses (Shanthi et al., 2025). AI's ability to analyze learning data in real-time allows lecturers and education managers to provide timely and specific interventions to students who need help (Pedro et al., 2019). In addition, AI also supports the concept of personalized learning. Through an intelligent recommendation system, AI can customize the content, speed, and style of material presentation according to the student's learning profile. Other research results proved that this kind of personalization not only improves learning outcomes, but also strengthens student engagement and motivation to continue learning (Sayed et al., 2023).

Transformative learning in higher education

Transformative learning is an approach in education that emphasizes the process of profound change in learners' perspectives, understanding, and behaviour (Illeris, 2014; Kurnia, 2021; Mezirow, 2018; Rojo et al., 2023). The concept was first introduced by Jack Mezirow in the late 1970s, who defined transformative learning as "the process of constructing meaning through critical reflection on personal assumptions, beliefs, and experiences" (Snyder, 2008). At the higher education level, transformative learning is seen as an important foundation in equipping students with critical thinking skills, independence, and adaptability amid the dynamics of global change (Hodge, 2014; Rojo et al., 2023).

The implementation of transformative learning in higher education is done through various learning strategies, such as reflective discussions, experiential projects, collaborative learning, and problem-based learning. A study conducted by Campbell (2017) found that student involvement in critical dialogue and reflective activities can encourage perspective transformation (Campbell & Brysiewicz, 2017). In addition, international experiences, internships, and community service activities have also proven effective in stimulating transformative learning (Briese et al., 2020). This approach provides space for students to challenge old assumptions and build new understandings that are more inclusive and contextualized.

The advantages of transformative learning in higher education have been proven through various studies (Kitchenham, 2008). Students who are involved in the transformative learning process tend to have a higher level of self-awareness, are able to solve problems creatively, and are better prepared to face the challenges of the world of work (Dirkx, 1998). Transformative learning also contributes to the formation of self-identity, increased empathy, and strengthening leadership competencies (Cranton, 2016; Illeris, 2014). In another study, Christie (2021) noted that transformative learning can increase student engagement in learning, thus having a positive impact on retention and academic achievement (Christie et al., 2015).

However, the implementation of transformative learning in higher education also faces various challenges. One of the main challenges is the resistance of students and lecturers to the change in learning paradigm that is more reflective and dialogical. Many students who are accustomed to conventional learning models find it difficult to actively engage in the process of critical reflection (Taylor, 2007). In addition, time constraints, heavy curriculum load, and lack of lecturer training in facilitating transformative learning are other obstacles that are often faced (Christie et al., 2015). Equally important, differences in students' cultures and backgrounds can also affect the effectiveness of transformative learning, especially in the increasingly multicultural context of higher education.

Methods

This research uses bibliometric analysis, a quantitative approach that is increasingly being adopted in scientific studies to measure, track and analyze literature trends in various fields of knowledge (Pessin et al., 2022). Through bibliometric analysis, researchers can identify the most influential publications, key authors, leading journals, as well as methodologies and research results that are the main references in a particular field (Rusydiana, 2021). In addition, the metadata of the articles analyzed provides a comprehensive picture of the development of a discipline, both in terms of publication volume, institutional collaboration, and a map of emerging topics (Lawani, 1981). This method has also been applied to examine various research topics, analyze the dynamics of journal publications, and map the contributions of various countries in the global scientific arena. Recent literature has also shown that bibliometric analysis can identify the rapid growth of publications on specific themes, such as innovative technological developments in education and healthcare, and uncover new trends that are gaining widespread attention in the scientific community (Zupic & Čater, 2015).

Collection and screening of data

The data collection and screening stage in this research analysis is very important because it can help in choosing the best bibliometric technique. Data for this study were extracted using the Scopus database-based search engine. With around 50 million literature items published since 1823, the Scopus database is the best for academic abstracts and citation counts (Thahirah et al., 2025). (TITLE-ABS-KEY (transformative learning) AND TITLE-ABS-KEY (higher education) AND TITLE-ABS-KEY (artificial intelligence)) AND PUBYEAR > 2018 AND PUBYEAR < 2026.

To ensure that this study focuses on the most relevant and recent research, specific inclusion criteria have been established. The decision to only include publications in English was made to standardize the language of analysis and ensure wider accessibility for an international audience. Restricting publications to the period between 2019 and 2025 aims to capture the most recent and relevant trends, reflecting the current state of artificial intelligence (AI) and transformative learning in higher education. By focusing solely on journal articles, this study seeks to analyze research that has undergone peer review, providing credible scientific insights on the topic. In addition, only publications that are in the final stages of the publication process are included to ensure that the articles analyzed reflect completed research with final data, thereby increasing the reliability and relevance of the findings. Grey literature, such as unpublished research reports or conference presentations, is not included in this study to maintain the quality and validity of the sources used. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement was used in this investigation to describe the study design. By using the PRISMA statement, authors can improve the reflection of systematic literature and meta-analyses (Sewell et al., 2023). Figure 1 displays the study design for the current period investigation, which used the PRISMA statement. The process of "identification, screening, eligibility, and inclusion" for records is also described in the diagram (Karimah & Hasegawa, 2022).

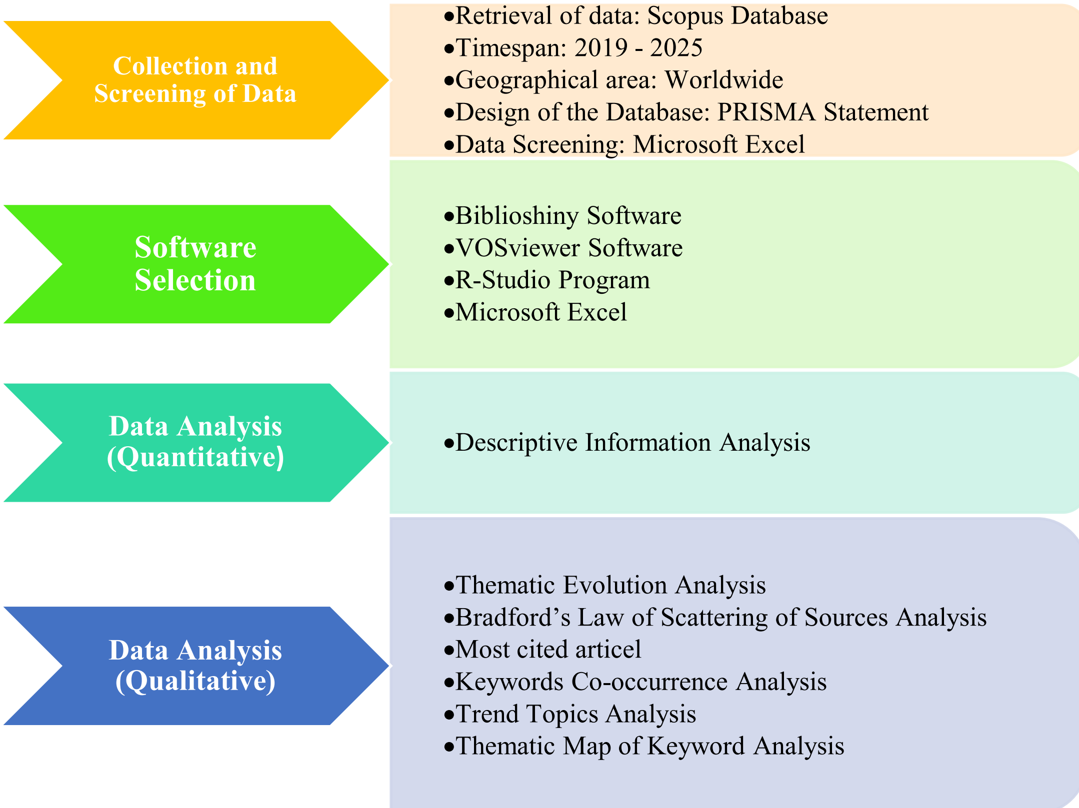

Figure 1: Summary of research method.

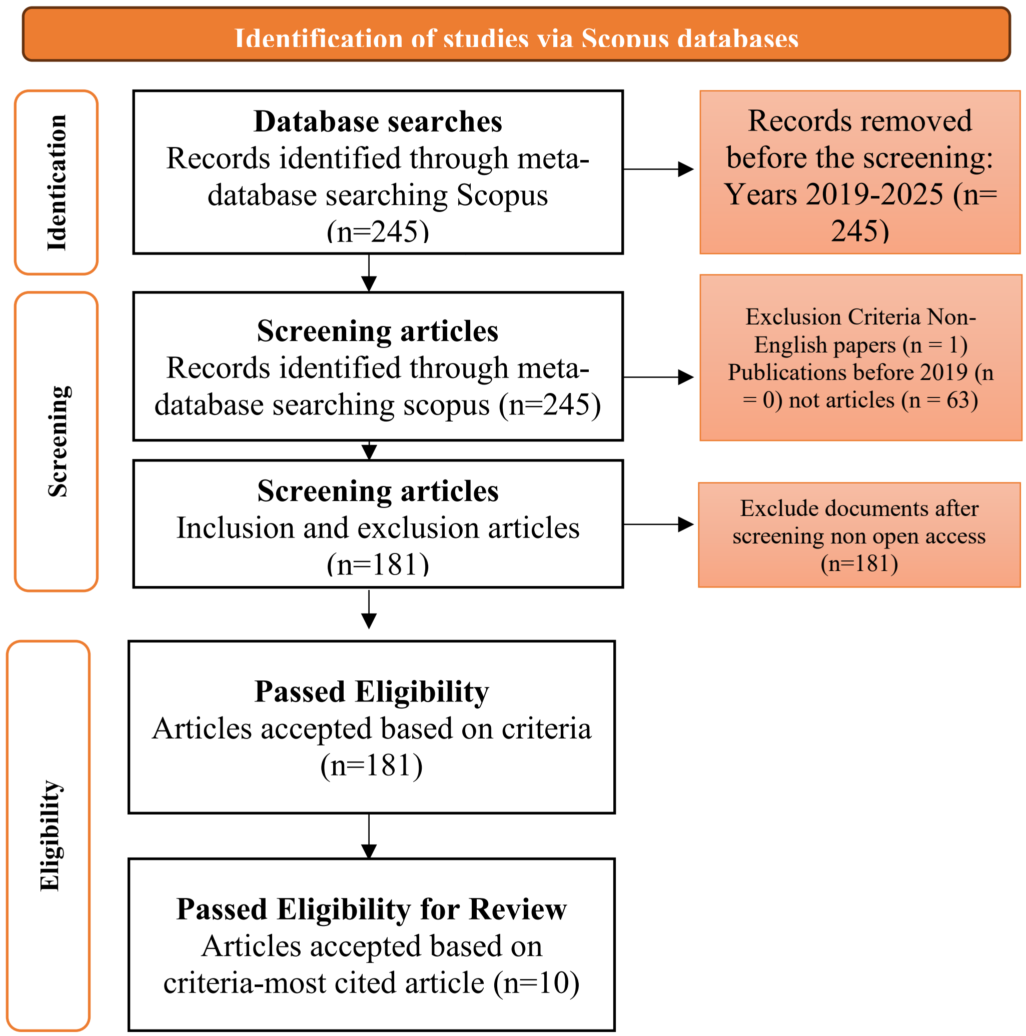

Figure 2: Prisma methods flow diagram.

Software selection

To perform a proper bibliometric analysis, appropriate software is required. Different types of analysis, such as content analysis, quantitative analysis, and visualization, can be carried out using a number of software and tools, such as "Bibliomatrix," "Vosviewer," "Gephi," "Citespace," "HistCite," "Publish or Perish," "Pajek," "Maxqda," "N Vivo," and "Bibexcel." In this study, the VOSviewer program and the Biblioshiny web application were used, along with the latest version of the BibliometricR package. The Bibliometrix R package, which is open-source software, was developed by Aria and Cuccurullo (2017) (Aria et al., 2024; Bhat et al., 2023). Compared to other software, Bibliometrics and VOSviewer offer faster analysis and generate data matrices to assess performance and provide scientific visualization of bibliographic databases (Bhat et al., 2023).

Data analysis

In this study, both quantitative and qualitative evaluations were taken into account in the analysis. The descriptive aspects of the refined empirical research included descriptive features, the most relevant and high-impact sources, the most cited articles, and the most relevant countries in the AI and transformative learning field, all of which were analyzed using Bibliometrix and SLR. The obtained data were then compiled in detail in CSV format by the researchers. During the data processing procedure, pre-analysis was also carried out using R-studio and Vosviewer software.

Findings and Discussion

The results and discussion of the study findings were carried out in this research. A detailed explanation is presented as follows:

RQ1. What are the descriptive features of the retrieved empirical studies and thematic evolution on AI and transformative learning?

Based on the data obtained in Figure 1 presents a comprehensive overview of bibliometric metrics on publications related to Artificial Intelligence (AI) and Transformative Learning in the context of Higher Education over the period 2019 to 2025.

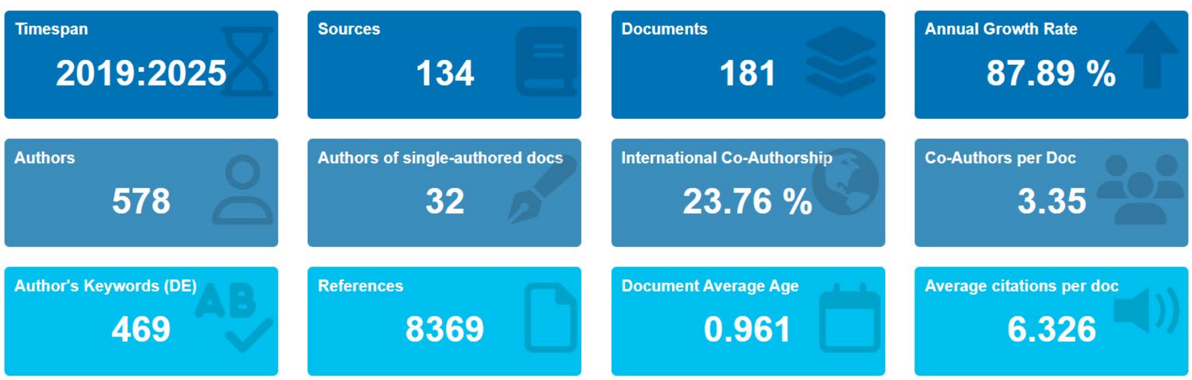

Figure 3: Descriptive features of the retrieved data.

Figure 1 shows the remarkable dynamism. A total of 181 documents coming from 134 different sources indicate that discussions and research on AI and transformative learning in higher education are spread across multiple platforms and journals. Most striking is the Annual Growth Rate of 87.89%, indicating a very rapid surge in interest and publication output. This reflects the relevance and urgency of this topic amidst the digital revolution and the need for innovative learning approaches.

In terms of contributors, 578 authors were working in this area. Interestingly, only 32 documents were the result of a single author, while the majority were collaborative works. The International Co-Authorship percentage of 23.76% underscores the global nature of the discussion, with researchers from different countries collaborating to explore the implications of AI and transformative learning in higher education. The average of 3.35 co-authors per document confirms that collaboration is the norm, perhaps due to the complexity of the topic requiring a wide range of expertise.

To provide an understanding of the development and trends of research in artificial intelligence (AI) and transformative learning over the period 2019-2025, the following thematic evolution map in Figure 3 visualizes the relationship between the author's country of origin, the author's name, and the main topics studied. This visualization provides a comprehensive picture of the geographical distribution, individual contributions, and dynamics of the evolving research themes from year to year.

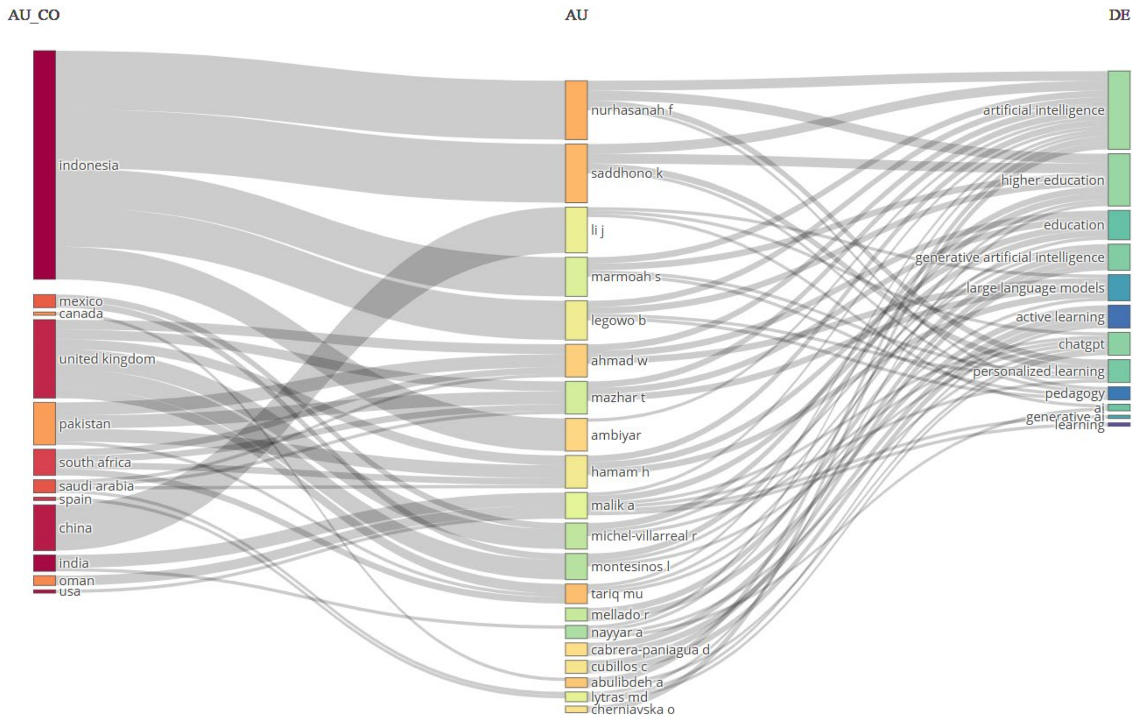

Figure 4: Thematic evolution map.

Figure 4 shows the strong representation of Indonesian researchers who contribute significantly across a range of themes, from artificial intelligence and higher education to generative AI and large language models. While there are contributions from other countries such as the UK, Mexico and China, the numbers are relatively small, highlighting the high concentration of research in Indonesia. Over time, research themes have shifted from general issues around AI in education to more specific topics, such as ChatGPT and generative AI, reflecting the rapid adoption of the latest technologies in education. Many Indonesian researchers are active in several themes at once, indicating a strong interdisciplinarity.

However, even though the technological aspect is more prominent, pedagogical themes such as personalized learning and active learning receive less attention, so the integration of AI in transformative learning still needs to be strengthened. Although AI can tailor educational content to individual needs, its role in encouraging active engagement through methods such as collaborative learning and critical reflection remains underexplored (Alyoussef et al., 2025). For AI to fully support transformative learning, it must be integrated into a pedagogical framework that encourages deeper engagement, critical thinking, and perspective transformation (Bartolomé et al., 2018; Vorobyeva et al., 2025; Walkington & Bernacki, 2020).

RQ2. What is the summary of related systematic literature reviews in AI and transformative learning?

To facilitate a deeper understanding of the research landscape, Table 1 provides an overview of significant systematic literature reviews in the context of artificial intelligence and transformative learning. Table 1 outlines the sources, timeframes, and main objectives of each review, thus enabling the identification of research directions and existing gaps. The fully analyzed article is included in Appendix A.

Table 1: Summary of related systematic literature reviews.

Systematic Literature Review Papers	**Source**	**Period**	**Objective**
Artificial Intelligence in Knowledge Management for Higher Education: Transformative Impact, Challenges, and Future Directions Post-COVID-19 (Hagos et al., 2025)	Articles	2020-2023	The objective of this study is to systematically review the application of Artificial Intelligence (AI) in Knowledge Management (KM) in higher education during the COVID-19 pandemic, as well as to analyze its effectiveness, challenges, and future research directions. This study also aims to identify research gaps and the need for a comprehensive ethical framework in the use of AI in education.
A Scoping Review of the Strategic Integration of Artificial Intelligence in Higher Education: Transforming University Excellence Themes and Strategic Planning in the Digital Era (Abulibdeh et al., 2025)	Articles	2000-2023	This research aims to explore the role of artificial intelligence (AI) in transforming higher education, particularly focusing on its impact on strategic development and achieving excellence in universities. The study aims to analyze the influence of AI tools on academic practices, student competencies, and institutional efficiency, particularly within the context of Qatar University.
The Promise and Pitfalls: A Literature Review of Generative Artificial Intelligence as a Learning Assistant in ICT Education (Chugh et al., 2025)	Articles	2022-2023	The objective of this research is to assess the efficacy and implications of integrating GenAI as a virtual learning assistant in ICT education. It aims to explore the tools, challenges, ethical considerations, and future directions for GenAI in the field.
Empowering learners with ChatGPT: insights from a systematic literature exploration (Mohebi, 2024)	Articles	2020-2024	The objective of this research is to conduct a systematic literature review on the integration of ChatGPT in academic settings. The objective of this study is to explore the challenges, factors driving its necessity, and the potential opportunities it offers in transforming academic practices. The study will focus on the adaptation and pedagogical integration of ChatGPT into educational institutions.

Prospers and Obstacles in Using Artificial Intelligence in Saudi Arabia Higher Education Institutions—The Potential of AI-Based Learning Outcomes (Alotaibi & Alshehri, 2023)	Articles	2011-2021	The objective of this research is to explore the opportunities and challenges related to the implementation of AI-based learning outcomes in higher education institutions in Saudi Arabia. Additionally, it aims to examine the contributions of major Saudi universities in advancing research on AI-based learning outcomes.
Artificial Intelligence in Higher Education: Trends, Possibilities and Challenges (Slavuj et al., 2024)	Conference paper	2023-2024	The objective of this research is to contribute to the broader understanding of the implications of AI adoption in higher education institutions and to inform current and future research directions and practical implementations in the field. The objective of this study is to provide a comprehensive review of the state-of-the-art in the application of artificial intelligence (AI) within the context of higher education. The review will entail an examination of contemporary trends, practical applications, and the prevailing challenges in the domains of teaching, learning, and administration.
Research Trends and Gaps in the Adoption of Immersive Reality Technologies in African Education Systems (Gudyanga, 2024)	Articles	2014-2023	The objective of this research is to analyze the research trends and gaps in the adoption of immersive reality technologies (IMRT) in African education systems. It aims to provide insights into the publication trends, research hotspots, collaboration networks, and funding patterns surrounding the use of immersive technologies like augmented reality (AR) and virtual reality (VR) in education across Africa. The study also seeks to identify research gaps and implications for future research, policy decisions, and enhancing the potential of immersive technologies in addressing educational challenges on the continent.
The Impact of ChatGPT on Student Learning Experience in Higher STEM Education: A Systematic Literature Review (Ilić et al., 2024)	Conference paper	2020-2024	The objective of this research is to systematically review the impact of ChatGPT on student learning experiences in higher STEM education. The objective of this study is to assess the impact of ChatGPT on student engagement, learning outcomes, and its effectiveness as a teaching tool. Additionally, the study will explore potential approaches for its application in the future.

Based on the systematic literature review presented, AI has become a major research focus in higher education, covering aspects ranging from knowledge management to strategic transformation of universities. These studies highlight the impact of AI, including GenAI and ChatGPT, as learning tools and examine the challenges and opportunities associated with its implementation in diverse academic environments, such as in Saudi Arabia and African education systems. The results show that AI has great potential to improve academic practices, student competencies, and institutional efficiency. However, there is also a focus on identifying research gaps, ethical considerations, and the need for a comprehensive framework to ensure responsible and effective implementation of AI in the future (Alahmed et al., 2023; Goktas & Grzybowski, 2025).

Most literature reviews on the integration of artificial intelligence (AI) in higher education are still limited to technical and managerial aspects, with little discussion of the pedagogical and ethical implications of its use. The impact of AI on knowledge management in higher education post-COVID-19, but this study emphasizes the effectiveness and challenges of the technology, while its pedagogical implications are still limited (Hagos et al., 2025). The role of AI in university strategic planning has been explored, but this research lacks discussion on how AI directly influences learning practices and student competency development (Abulibdeh et al., 2025). This indicates that further research is needed to explore how AI can support more inclusive and transformative learning, as well as to understand the ethical challenges associated with its use in the classroom (Chugh et al., 2025).

Furthermore, despite the positive impact of ChatGPT on learning, the study does not sufficiently discuss the risk of dependence on AI, which can reduce students' critical thinking skills and weaken social interaction among learners (Mohebi, 2024). Further research should explore in greater depth how AI can be effectively integrated into the curriculum to support reflective and collaborative learning, rather than merely as a tool that accelerates individual learning (Slavuj et al., 2024). In addition, other studies show the potential of AI in improving learning outcomes in Saudi Arabia, but a more comprehensive approach is still needed regarding the role of AI in strengthening students' critical and analytical thinking skills in the context of transformative learning (Alotaibi & Alshehri, 2023).

RQ3. What are the most relevant high-impact sources through Bradford's Law of Scattering in the field of AI and transformative learning?

Bradford's Law measures the gradual decrease in benefits when searching for references in scientific publications (Gupta et al., 2024; Villalobos et al., 2022). This law explains how information is spread out in publications covering a particular topic. For example, if journals in the field are divided into three groups based on the number of studies published, each group will hold about one-third of the total publications, with calculations showing that the number of publications in each category is $1:n:n^2$ (Bradford, 1934).

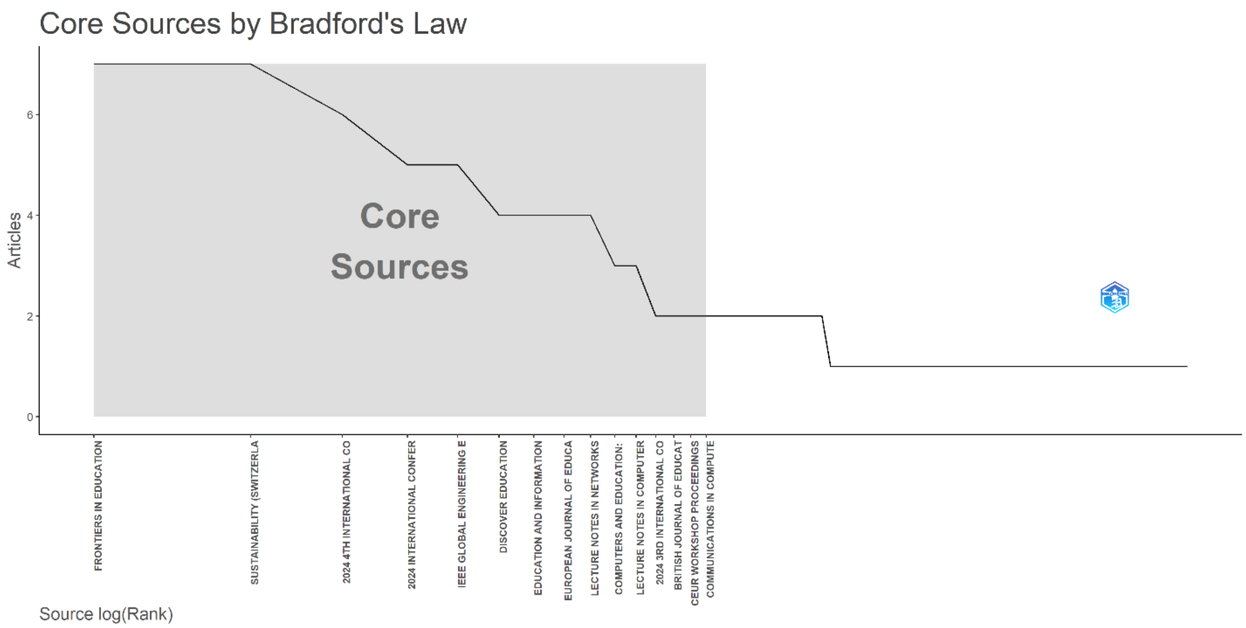

Figure 5: Core sources by Bradford's Law.

Figure 5 represents the application of Bradford's Law to identify core sources in a particular research field, which in this case relates to education and technology. Based on the visualization, it can be seen that a number of journals, such as Frontiers in Education, Sustainability (Switzerland), and IEEE Global Engineering Education Conference, occupy the topmost position as the main sources with the most article contributions. The shaded area indicates the core sources zone, which signifies the most productive group of journals or proceedings that play a central role in distributing knowledge in this field. The distribution of the number of articles on the vertical axis shows a sharp decline after the core zone, according to the pattern described by Bradford's Law, where a few sources contribute many publications, while the majority of other sources contribute only a few publications.

While the identification of core sources is helpful in prioritizing literature, there is a risk of bias that can occur when relying solely on core sources. The tendency to refer to popular or frequently indexed journals can lead to the exclusion of innovative research that may be published in non-core journals or new journals that have not received many citations but have the potential to contribute greatly to the development of science.

RQ4. What are the most cited articles on AI and transformative learning?

To provide a more comprehensive overview of the most influential articles in the field of AI and transformative learning, a summary of the documents with the highest number of global citations is presented below. Each article is presented by code, title, year, publication source, total citations, and key findings identified in the study. The details can be seen in Table 2.

Table 2**:** Documents with the highest global citations in the dataset.

Article Code	Title (Author, Year)	Source	Total citations	Important findings
A1	Navigating the confluence of artificial intelligence and education for sustainable development in the era of industry 4.0: Challenges, opportunities, and ethical dimensions (Abulibdeh et al., 2024)	Journal of Cleaner Production	190	Particularly ChatGPT, artificial intelligence tools are changing the educational scene and supporting curriculum redesign, individualized learning, and creative output. There are moral issues needing attention.
A2	Exploring Artificial Intelligence in Academic Essay: Higher Education Student's Perspective (Malik et al., 2023)	International Journal of Educational Research Open	116	Writing tools driven by artificial intelligence improve students' academic performance by means of grammar, plagiarism detection, and general writing quality enhancement. Concerns about critical thinking and creativity surface and demand careful integration of artificial intelligence.
A3	Artificial intelligence in language instruction: impact on English learning achievement, L2 motivation, and self-regulated learning (Malik et al., 2023)	Frontiers in Psychology	96	AI in language learning boosts achievement, motivation, and self-regulated learning, with significant improvements in grammar, vocabulary, and writing skills
A4	Drivers and Consequences of ChatGPT Use in Higher Education: Key Stakeholder Perspectives (Hasanein & Sobaih, 2023)	European Journal of Investigation in Health, Psychology and Education	84	Including ChatGPT into higher education provides tailored learning, so enhancing student information access and study techniques. It also begs questions about academic integrity and too heavy reliance on artificial intelligence tools.
A5	Prospers and Obstacles in Using Artificial Intelligence in Saudi	Sustainability (Switzerland)	56	AI adoption in Saudi Arabian universities presents significant

	Arabia Higher Education Institutions—The Potential of AI-Based Learning Outcomes (Alotaibi & Alshehri, 2023)			potential for educational improvement but faces challenges like the need for faculty training and infrastructure development
A6	Using Generative Artificial Intelligence Tools to Explain and Enhance Experiential Learning for Authentic Assessment (Salinas-Navarro et al., 2024)	Education Sciences	55	Generative AI tools are transforming experiential learning by enhancing authentic assessments and fostering higher-order thinking
A7	Effects of Generative Chatbots in Higher Education (Akmam et al., 2023)	Information (Switzerland)	54	Chatbots in education facilitate interaction, provide personalized feedback, and reduce instructor workload, but challenges in accuracy and over-reliance need to be addressed
A8	The influence of sociodemographic factors on students' attitudes toward AI-generated video content creation (Pellas, 2023)	Smart Learning Environments	34	Students' opinions of AI-generated video content are much influenced by sociodemographic elements, which also affect their readiness to use AI tools for learning.
A9	Accelerated chemical science with AI (Back et al., 2024)	Digital Discovery	34	AI is revolutionizing chemical research by accelerating the discovery process, helping to address complex problems in fields like renewable energy
A10	Problematising Artificial Intelligence in Social Work Education: Challenges, Issues and Possibilities (Hodgson et al., 2022)	British Journal of Social Work	24	AI integration in social work education presents both challenges and opportunities, particularly in terms of ethical concerns and the need for curriculum adjustments

Table 2 highlights the top ten globally cited articles in systematic reviews of artificial intelligence (AI) in education. The top-ranked article is (Abulibdeh et al., 2024) published in the Journal of Cleaner Production, with 190 global citations. The main findings show that AI, especially ChatGPT, has brought significant changes to the educational landscape by supporting curriculum redesign, personalized learning, and enhancing creativity. However, ethical aspects are an important concern that must be carefully managed in the implementation of AI in education. Various other articles on the list also reveal the dynamics of the role of AI in supporting the teaching and learning process. How AI-based writing aids can improve students' academic performance, from detecting plagiarism to improving the quality of writing (Malik et al, 2023). Meanwhile, other research underlines that the integration of ChatGPT in higher education provides opportunities for learning personalization, but on the other hand, raises concerns about academic integrity and the potential for over-reliance on AI (Hasanein & Sobaih, 2023). The diversity of findings from each article shows that AI not only has an impact on the technical aspects of learning, but also on the dimensions of motivation, self-regulation, and the development of higher-order thinking skills.

In the context of transformative learning, the integration of AI in education presents both challenges and opportunities. Transformative(Peters & Göhlich, 2024) learning focuses on developing critical thinking and personal growth through experiences that challenge existing beliefs and assumptions. The impact of AI in education, as highlighted by (Flores-Viva & García-Peñalvo, 2023; Nedungadi et al., 2024) aligns with this by offering the potential to revolutionize how students acquire knowledge, encouraging them to adapt and rethink their learning processes. However, for AI to effectively support transformative learning, educators must address ethical issues, ensure proper training, and build infrastructure that supports this change, while maintaining the originality and critical thinking that are at the heart of transformative education. The challenge lies in balancing AI's capacity to enhance learning with the need to maintain human-centred, reflective, and knowledge-based practices that foster deep and transformative learning experiences. Therefore, appropriate regulation, continuous capacity building, and further study are essential to ensure that the application of AI in education supports, rather than hinders, the transformative learning process.

RQ5. What are the most relevant** authors' **countries for AI and transformative learning?

To identify the countries that have made the greatest contribution to this concept, it is important to sort key publications (Liñán & Fayolle, 2015) on AI and transformative learning by geographical region. By conducting a more accurate comparison, a contextual evaluation can help infer relevant research gaps for further study, as well as improve researchers' understanding of AI and transformative learning at the national level. The scientific output of the author countries can be seen in Figure 6.

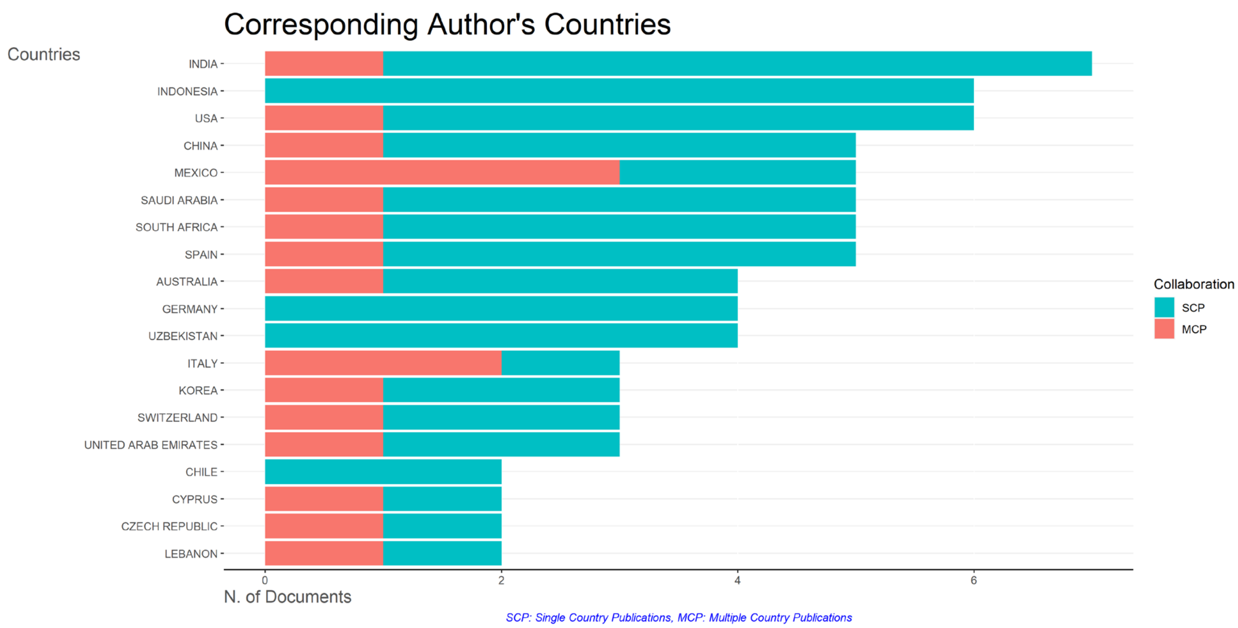

Figure 6: Corresponding author’s countries.

Based on Figure 6, the distribution of countries of corresponding authors in scientific publications, it can be interpreted that India is the country with the highest contribution in terms of the number of documents, followed by Indonesia, the United States, and China. This graph also distinguishes between publications written by authors from only one country (Single Country Publications/SCP) and publications resulting from cross-country collaboration (Multiple Country Publications/MCP). In general, most of the documents come from SCP, which can be seen from the dominance of the blue colour in the graph. This shows that research in this field is still dominated by internal collaboration in one country. However, there are also contributions from MCP in several countries such as the United States, China, and Mexico, which indicates international collaboration, although the portion is still smaller than SCP.

Interestingly, India and Indonesia not only occupy the top positions in terms of the number of publications, but also show a pattern of collaboration that still tends to be national. Meanwhile, countries such as the United States and China are starting to show a trend of involvement in international collaboration (MCP). This finding indicates that, although cross-country collaboration is starting to grow, the main contribution of scientific publications in the observed topics still comes from national collaboration. This can be a basis for researchers and policymakers to encourage increased international collaboration to expand scientific networks and increase research impact.

One major collaboration was a study written by authors from several countries and involving four countries, such as China, Japan, Mongolia, and the United States, as research samples. This study discussed Generative AI policies in higher education using the Qualitative Comparative Analysis (QCA) method. The findings show that Japan and the United States emphasize a human-centred approach, with the aim of supporting educators and improving teaching methods, while China and Mongolia prioritize national security and digital progress, with less emphasis on educational applications. Although all countries focus on diversity, equity, and inclusion, this study also highlights the lack of attention to the digital divide in the implementation of AI in education(Xie et al., 2024).

RQ6. What are the trending research themes for future investigations in the field of AI and transformative learning?

Keywords are essential indicators of research focus, topics, trends, and development directions because they summarize an article's main concepts and essence (Liñán & Fayolle, 2015; Wang & Chai, 2018; Yuan et al., 2022). Keyword analysis, especially with keyword clustering and keyword density visualization using VOSviewer, helps identify significant trends in a research field (Bukar et al., 2023). The primary purpose of keyword co-occurrence analysis is to evaluate the relationships between keywords in articles to find popular topics and improve academics' understanding of ongoing scientific issues in AI and transformative learning.

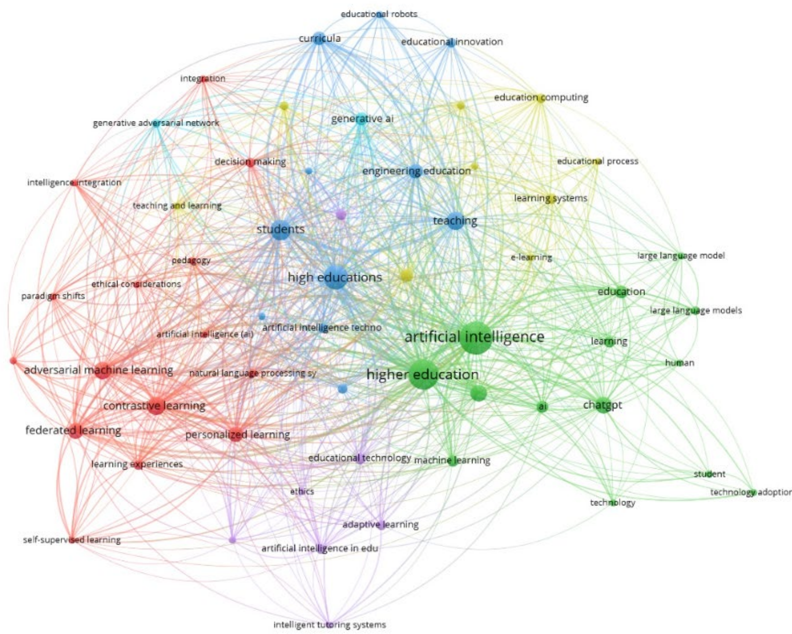

Figure 7: Network visualization.

Figure 7 presents a bibliometric network visualization that maps the relationship between research topics on artificial intelligence (AI) in higher education. Through the VOSviewer approach, it can be seen that research centres in this field focus on keywords such as artificial intelligence, higher education, and learning. These keywords appear dominant in larger node sizes and become the center of connection for other themes around them.

The network shows that AI research in higher education does not stand alone, but forms a tightly connected thematic network. Topics such as ChatGPT, large language models, personalized learning, federated learning, and adversarial machine learning show that current research trends are moving towards the use of generative AI, personalized learning technologies, and the development of intelligent systems that are increasingly adaptive to student needs. In addition, the integration of keywords such as ethics, educational technology, and curricula emphasizes the importance of ethical issues, educational technology innovation, and AI-based curriculum development in higher education. This visualization confirms that AI research in higher education is multidisciplinary and dynamically evolving. The complexity of interdisciplinary relationships reflects the evolving challenges and opportunities in terms of technological, pedagogical, and ethical aspects. Thus, thematic mapping such as this is an important foundation for identifying research gaps, strengthening cross-field collaboration, and deepening critical studies related to the application of AI in higher education, especially in the era of the emergence of new technologies such as ChatGPT and large language models.

Figure 8 is a tree map showing the distribution of various topics or themes frequently appearing in research or literature. Each box represents a particular theme, and the size of the box reflects the proportion or frequency of occurrence of that theme.

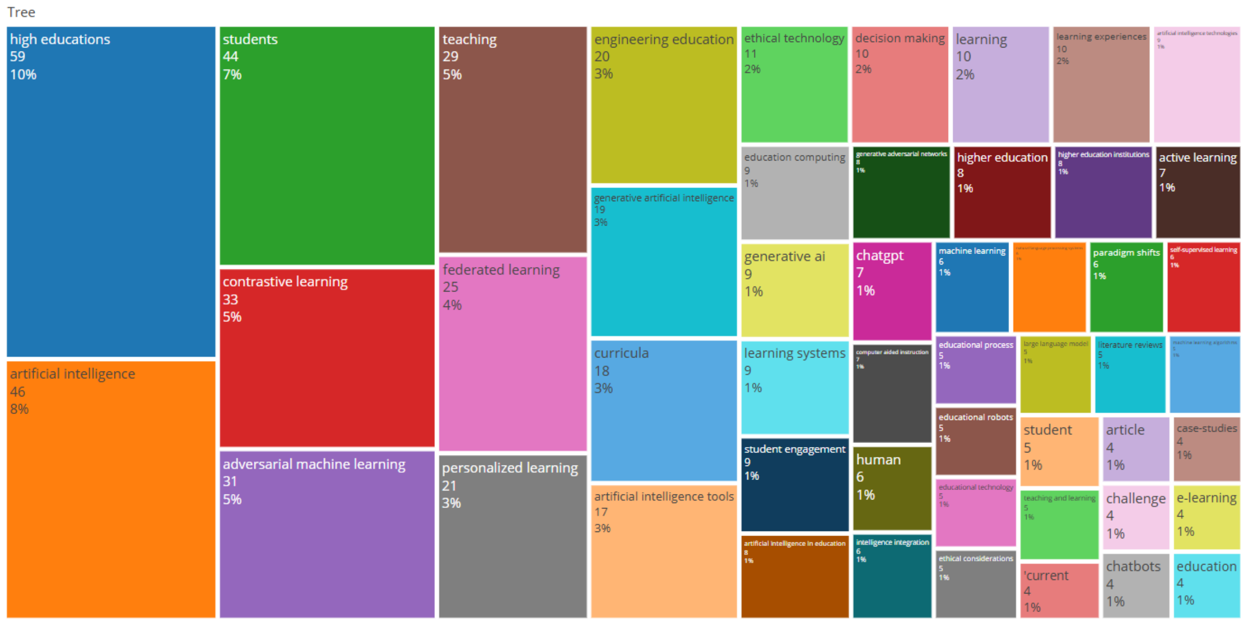

Figure 8: Tree map.

The tree map shown represents the distribution of the main keywords in research on artificial intelligence in higher education. From this visualization, it can be seen that the topics of “higher education” and “artificial intelligence” dominate the discourse, confirming that the integration of AI in higher education environments is a central concern in recent research. Other keywords such as “students”, ‘teaching’, and “learning” also occupy a large proportion, indicating a strong focus on the role of students and the learning and teaching process in the context of AI adoption.

In addition to these core themes, many studies also highlight new approaches to AI-based learning, reflected by the emergence of terms such as “contrastive learning”, “federated learning”, “adversarial machine learning”, and “personalized learning”. The presence of these buzzwords indicates that innovation in learning technology is one of the main directions of development, where AI is expected to provide learning that is increasingly adaptive and responsive to individual needs. However, some important issues such as “ethics”, “student engagement”, and “paradigm shifts” appear to appear with less frequency. Yet, these issues are crucial to ensure the responsible and sustainable application of AI in education. This indicates the need to expand the scope of research to social, ethical, and educational paradigm shifts, so that the utilization of AI in higher education is not only oriented towards technological efficiency, but also human values and strengthening student character (Al-Zahrani & Alasmari, 2024; Bucea-Manea-Țoniş et al., 2022; George & Wooden, 2023).

Furthermore, upon further analysis, the findings in one of the previous studies (Agarwal et al., 2025)are in line with (Illeris, 2014), which emphasizes the importance of profound changes in educational perspectives and practices through critical reflection. In the higher education sector in

India, the integration of AI and Education 5.0 requires educators to reflect on and change their traditional teaching methods. Identified challenges, such as resistance to change and lack of training, hinder this process. However, through targeted policies and adequate training programs, universities can facilitate a transformative learning process that enables educators to adopt new educational technologies. This research contributes to the understanding of the relationship between transformative learning and technological empowerment in education, providing insights relevant to similar global contexts.

Conclusion

Based on research conducted on Artificial Intelligence (AI) and transformative learning in higher education, it can be concluded that (RQ1) research on AI and transformative learning in higher education shows a significant increase in the number of publications from 2019 to 2025. In general, the research theme began with general issues regarding the application of AI in education, but over time, it has developed into more specific topics, such as generative AI and personalized learning that support transformative learning; (RQ2) the systematic literature reviews (SLRs) reveal that AI plays a transformative role in higher education, improving personalized learning and knowledge management. However, the SLRs also highlight challenges, including the need for comprehensive ethical frameworks and the potential risks of AI over-dependence. Despite the promising applications of tools like ChatGPT, there is still a lack of discussion on the pedagogical implications and the integration of AI in transformative learning practices; (RQ3) using Bradford's Law, it was identified that several major journals, such as Frontiers in Education and Sustainability (Switzerland) dominate publications in the field of AI and transformative learning. These journals play a central role in distributing knowledge related to the integration of AI in higher education. However, there is still room for more innovative research to be published in new journals that are not indexed in the core group. This highlights the importance of covering more literature to see broader developments in the application of AI in the context of transformative education (RQ4). The most cited articles in this field show that AI is increasingly accepted in transformative learning, with applications such as ChatGPT and AI in authentic assessment helping to improve student engagement and personalized learning. However, issues of over-reliance on technology and ethical challenges related to academic integrity remain a major focus. This reminds us of the importance of balancing technology with the development of critical thinking skills and creativity in the learning process (RQ5). India and Indonesia are the countries with the largest contributions to publications related to AI and transformative learning, with a predominance of national collaborations. Although there is little contribution from international collaboration, countries such as the United States and China are beginning to show a tendency to strengthen cooperation between countries. This emphasizes the importance of enhancing international collaboration to expand research networks and the impact of AI application in transformative education (RQ6). Some of the trending research themes in the field of AI and transformative learning are personalized learning, generative AI, and intelligent learning systems that are increasingly responsive to the individual needs of students. This trend highlights the importance of AI in improving a more adaptive learning experience that is in line with the development of transformative learning. However, issues of ethics, educational paradigms, and student engagement in the application of AI as a learning tool still need more attention in future research.

Limitations

This study has several limitations that need to be noted. First, the literature search was limited to the Scopus database and English-language articles published in 2019-2025, so there may be relevant research in other languages or gray literature that was not reached. Second, while bibliometric analysis offers quantitative insights, it does not fully capture the depth and contextual nuances of qualitative research on transformative learning and AI integration. Third, the dominance of national collaborations and contributions from specific countries, particularly Indonesia and India, has the potential to create geographical biases that limit the generalizability of the results. In addition, the rapid development of AI technology means that some recent innovations may not be optimally represented in the academic literature. Fourth, the study does not incorporate primary empirical data or conduct a meta-analytic synthesis, which limits the ability to assess effect sizes and compare outcomes across studies in a statistically rigorous manner.

Suggestions for future research

Future research should emphasize cross-national and multidisciplinary collaboration to enrich the understanding of AI-based transformative learning in various educational and cultural contexts. In addition, future research should expand its geographical scope to include more publications from other countries and regions by searching for information from other publication databases to gain a more comprehensive and representative understanding of AI integration in transformative learning in various global contexts. Researchers are advised to expand the scope of bibliometric analysis to include other databases and non-English language sources for a more representative global perspective. In addition, in-depth qualitative studies are needed to explore the experiences of students and faculty in AI-based learning environments. Ethical, social, and policy issues related to AI adoption-including protection of academic integrity, enhancement of digital literacy, and inclusivity for vulnerable groups-should be high on the agenda of future research. The development and evaluation of holistic frameworks that combine technological innovation with critical pedagogy, ethics, and student empowerment are needed to make transformative learning in the AI era more relevant and sustainable.

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